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Instruction/Notes: Solve All Questions

Q.1 We have the following program segment executed on a three stage (I, E, D) RISC pipeline. The pipeline allows two simultaneous memory accesses. [PLO 1] [CLO 5] [20]

100 LOAD R1, X

101 LOAD R2, Y

102 INC R1 (2)

103 DEC R2 (2)

104 BRANCH 107

105 ADD R1, 5

106 SUB R2, 2

107 STORE R2, Z

Load from Address X

Load from Address Y

Increment R1

Decrement R2

Branch to Address 107

Add 5 to R1

Subtract 2 from R2

Store R2 at Address Z

Using delayed branch in pipeline operation in the above program, draw the pipeline timing diagrams:

- i- using NOOPs 10
- ii- after rearranging the instructions (optimized delayed branch) 9

Calculate the clock cycles required to execute the above program in each case.

Q.2 Assume the following data (in decimal) is stored in the memory and processor registers.

Memory Location 200 contains 400, Memory Location 300 contains 500,

Memory Location 400 contains 600, Memory Location 500 contains 700,

REGISTER R1 contains 300, REGISTER R2 contains 500

After executing each of the following machine instructions, what values will be loaded into the register AC?

[PLO 1] [CLO 4] [20]

- i- LOAD AC IMMEDIATE 400
- ii- LOAD AC DIRECT 300
- iii- LOAD AC INDIRECT 200
- iv- LOAD AC REGISTER R1
- v- LOAD AC REGISTER INDIRECT R2

Q.3 a To add two floating point numbers you must adjust the exponents (by shifting the fraction) to make them the same. Then you can add the fractions and normalize the result, if it needed be. Add the single precision IEEE numbers 3EE00000H and ED800000H and express the normalized result in hexadecimal.

92% b. Convert 12.125 to IEEE single precision floating point format. Give the result as eight hexadecimal digits. 4142 [CLO 3] [10 + 10]

Q.4 a. A set associative cache memory has a line size of eight bytes and a set size of four. The cache can accommodate a total of 65536 bytes. The main memory size that is cacheable is $4M \times 8$ bits and a block size of eight bytes. Design the cache structure, and show how the processor's addresses are interpreted?

b. How many $1M \times 8$ RAM chips are needed to construct a memory capacity of $32M \times 16$? How many lines of the address bus must be used to access $32M$ words of memory? How many of these lines will be common to all chips? How many lines must be decoded for chip select? Specify the size of decoder? $2^3 \rightarrow 2^9$ [CLO 2] [10 + 10]

Q.5 For the following Two Hamming code words X and Y, test if these code words are correct, assuming they were created using an even parity Hamming Code with left most bit being the MSB. If one is incorrect, indicate what the correct code word should have been. Also, indicate what the original data was. Assume that no more than one bit is in error. [CLO 2] [20]

$$X = 010101100011$$

$$Y = 111110001100 - 2^2$$

$$\begin{array}{l} 2^0 - C1 \\ 2^1 - C2 \\ 2^2 - C4 \\ 2^3 - C8 \end{array}$$

R2

R1

$$-1 \times 2^{-1} = 126$$